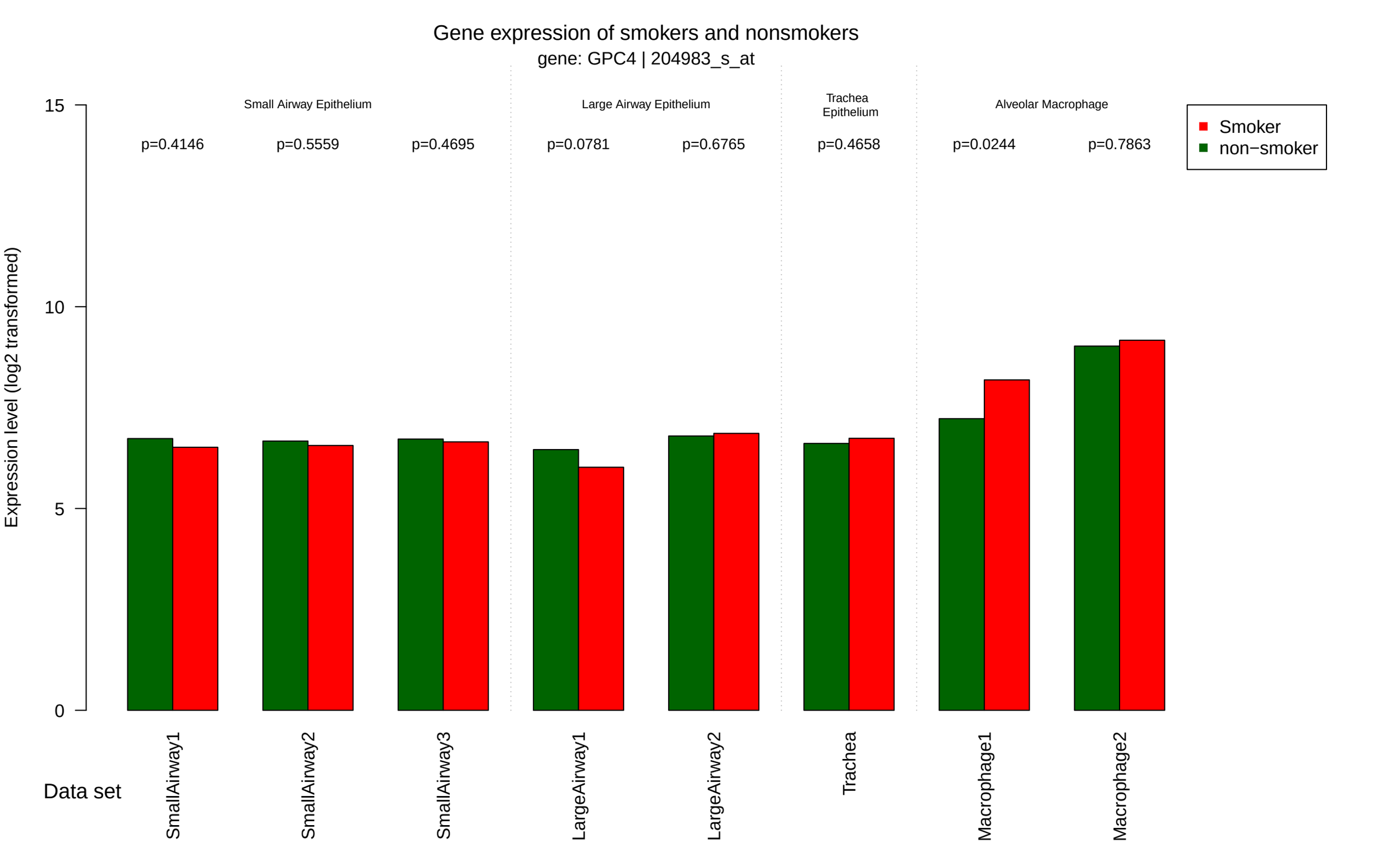
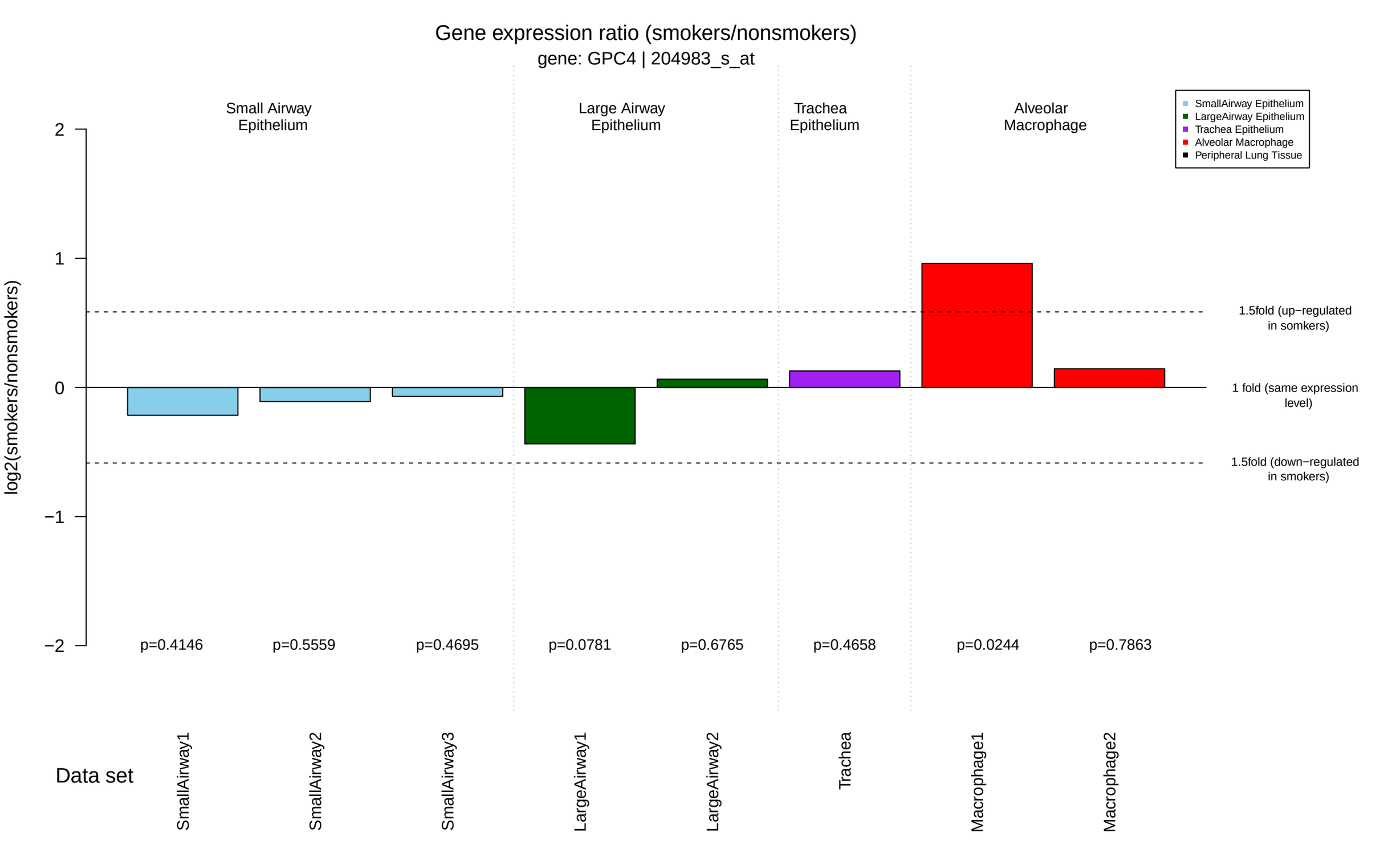
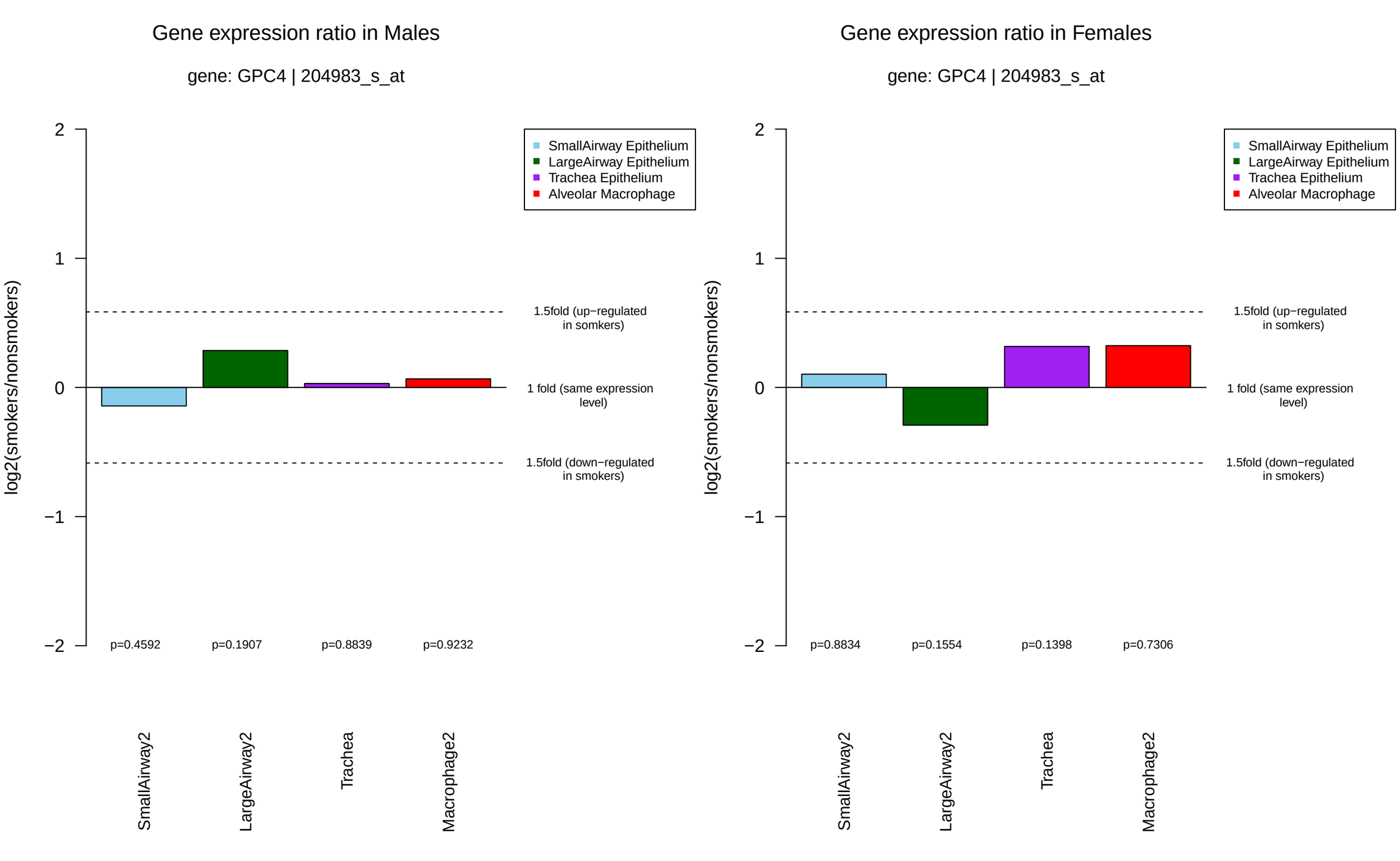
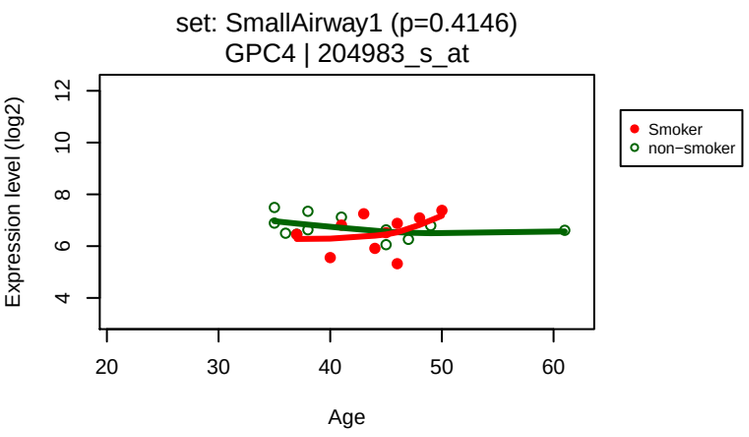
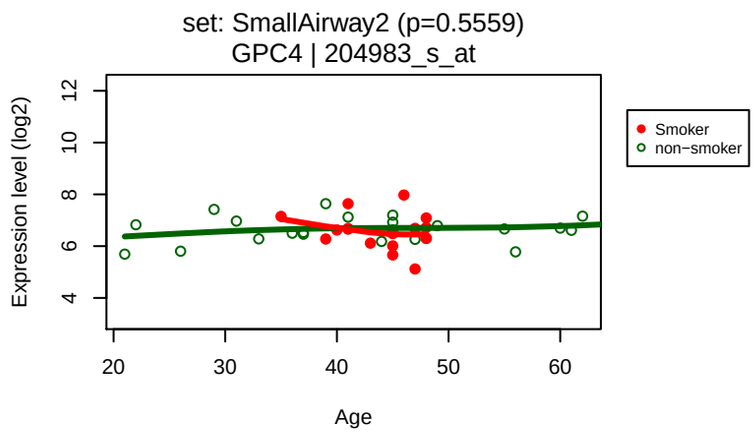
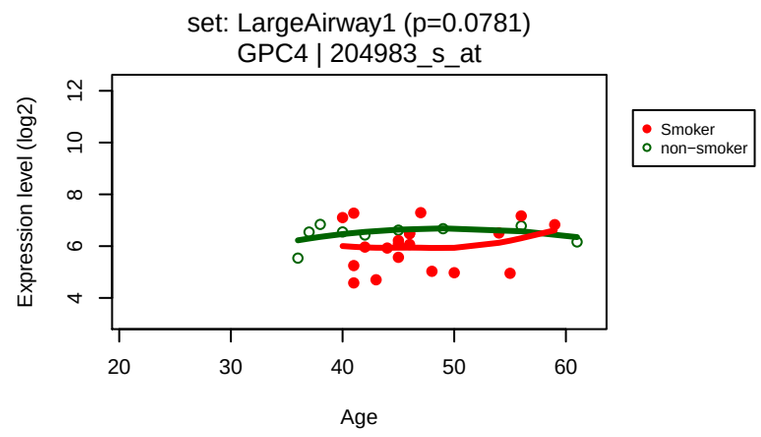

Gene expression of smokers and nonsmokers
gene: GPC4 | 204983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4146
p=0.5559
p=0.4695
p=0.0781
p=0.6765
p=0.4658
p=0.0244
p=0.7863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

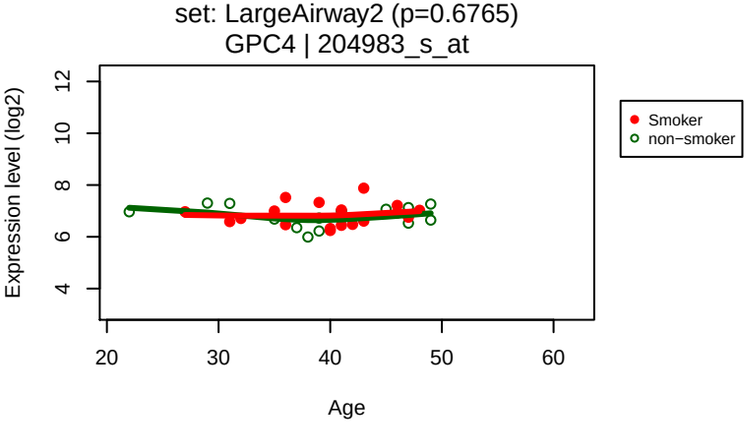

Gene expression ratio (smokers/nonsmokers)
gene: GPC4 | 204983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4146
p=0.5559
p=0.4695
p=0.0781
p=0.6765
p=0.4658
p=0.0244
p=0.7863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

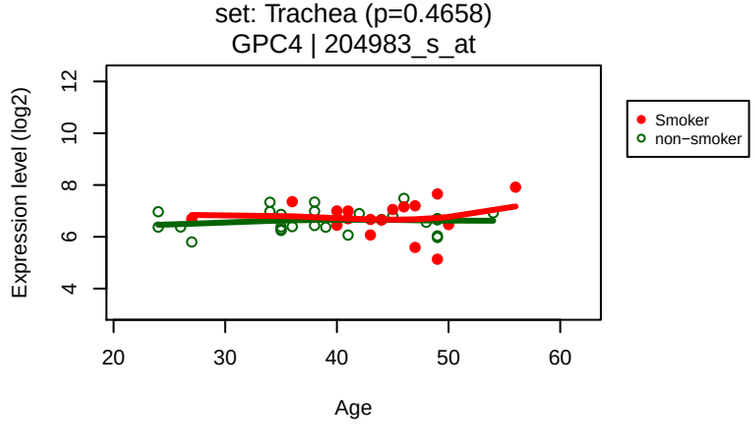

Gene expression ratio in Males
gene: GPC4 | 204983_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4592
p=0.1907
p=0.8839
p=0.9232
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPC4 | 204983_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8834
p=0.1554
p=0.1398
p=0.7306
SmallAirway2
LargeAirway2
Trachea
Macrophage2

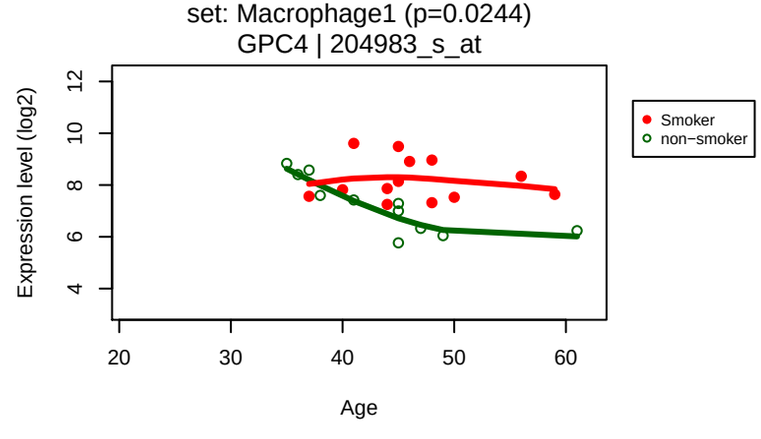

set: SmallAirway1 (p=0.4146)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker

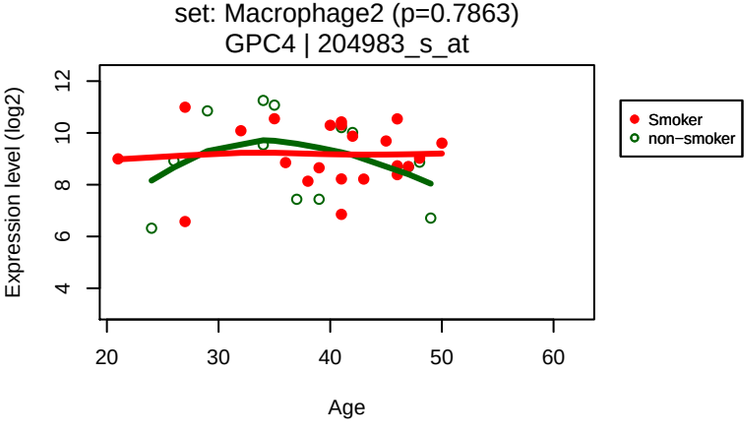

set: SmallAirway2 (p=0.5559)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker

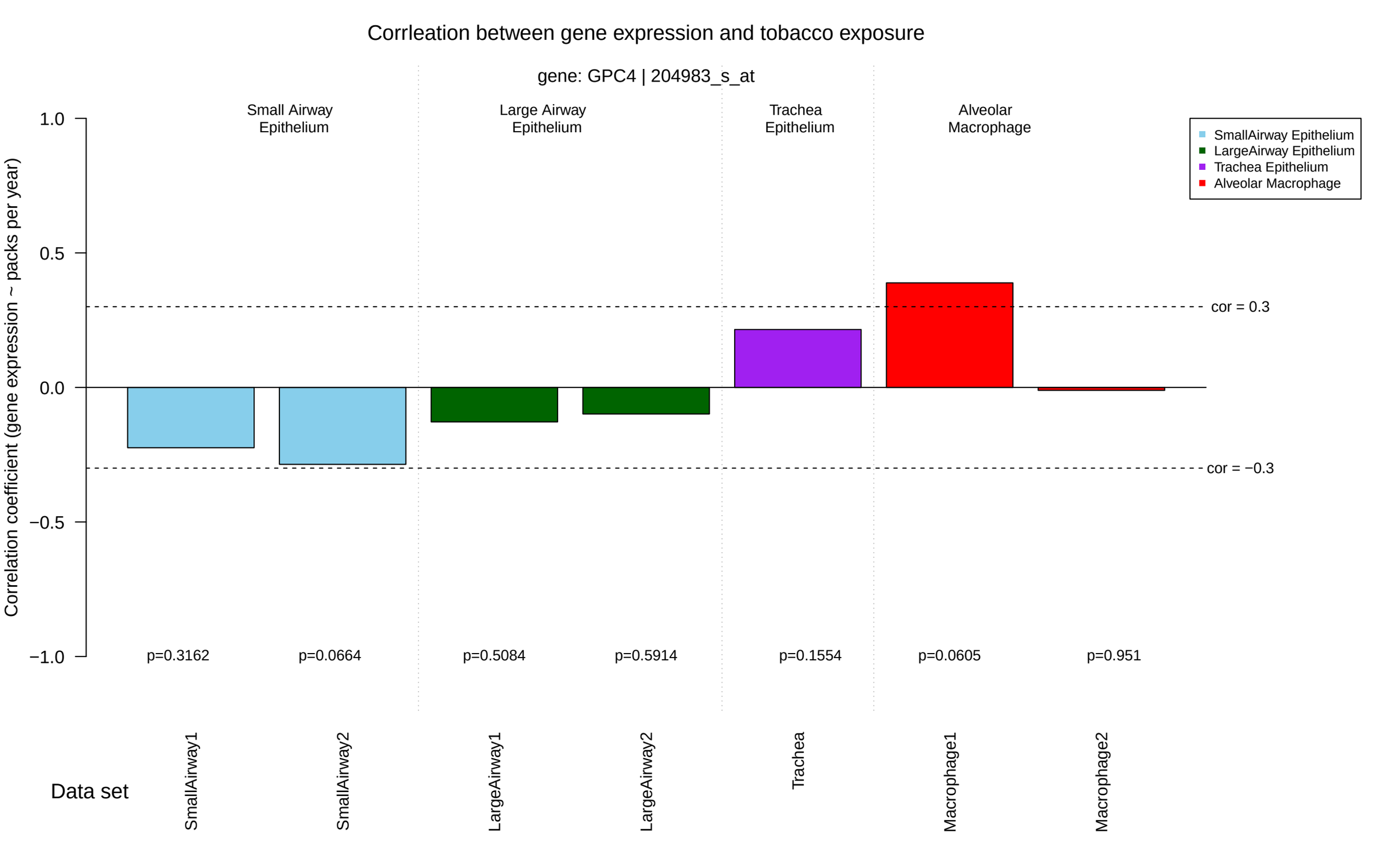

set: LargeAirway1 (p=0.0781)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6765)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4658)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0244)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7863)
GPC4 | 204983_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPC4 | 204983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3162
p=0.0664
p=0.5084
p=0.5914
p=0.1554
p=0.0605
p=0.951
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2